LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	28
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Functions of signal processing elements in a FCS, signal processing channel and its elements, limiting synchronizing, gain and adaptive control
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Functions of signal processing elements in a FCS, signal processing channel and its elements, limiting synchronizing, gain and adaptive control, feedback in a FCS.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of signal processing block diagrams and identification of limiting/synchronizing functions in provided schematics.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an AFCS?
(b) Define the term 'signal processing' in the context of flight controls.
(c) List three common elements found in a signal processing channel.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Signal processing channel elements • Function of limiting and synchronizing • Gain control mechanisms • Adaptive control principles • Feedback loop integration	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of limiting circuits • Synchronization techniques in FCS • Gain scheduling applications • Adaptive control algorithms • Feedback stability criteria	L & Ds	PPT, PDF	20 min
(c)	• Integration of signal processing in FCS • Impact of adaptive control on performance • Troubleshooting signal processing faults • Case studies of FCS failures • Future trends in signal processing	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how limiting and synchronizing functions prevent control surface saturation.
(b) Describe the role of adaptive control in varying flight conditions.
(c) Identify the key components of a feedback loop in an AFCS.

12. Home Assignment and Follow Up	Prepare a brief note on the purpose and explanation of altitude hold and airspeed hold systems as covered in LP 29.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Purpose and explanation of altitude hold and airspeed hold.	Refer to the master syllabus reference list	LP 29

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min